



***satu*itb**

150²⁰30

**DIKTISAINTEK
BERDAMPAK**

FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM

INSTITUT TEKNOLOGI BANDUNG



FMIPA ITB

Fakultas Matematika dan Ilmu Pengetahuan Alam (FMIPA) merupakan fakultas tertua dan terdepan di Indonesia dalam bidang matematika dan ilmu pengetahuan alam. Dibentuk pada tanggal **6 Oktober 1947** dengan sebutan *Faculteit van Exacte Wetenschap*. Pada tanggal **22 September 1948** nama fakultas ini berubah menjadi *Faculteit van Wiskunde en Natuur Wetenschap*. Di Tahun **1950** berubah menjadi Fakultas Ilmu Pengetahuan dan Ilmu Alam (FIPIA), dan pada tahun **1972** hingga sekarang berubah menjadi Fakultas Matematika dan Ilmu Pengetahuan Alam (FMIPA).



STATUS AKREDITASI PROGRAM STUDI SARJANA FMIPA

Program Studi	Akreditasi	No. SK	Akreditasi Internasional
Matematika	Unggul	008/SK/LAMSAMA/Akred/S/II/2026	 ASIIN
Fisika	Unggul	12615/SK/BAN-PT/Ak-PPJ/S/XI/2021	 ASIIN
Astronomi	Unggul	14116/SK/BAN-PT/Akred-PMT/S/III/2022	 ASIIN
Kimia	Unggul	003/SK/LAMSAMA/K-Akred/S/III/2023	 ROYAL SOCIETY OF CHEMISTRY
Aktuaria	Unggul	191/SK/LAMSAMA/Akred/S/VII/2025	



Accredited & Certified by:



STATUS AKREDITASI PROGRAM STUDI MAGISTER FMIPA

Program Studi	Akreditasi	No. SK
Matematika	Unggul	180/SK/LAMSAMA/Akred/M/VI/2025
Fisika	Unggul	099/SK/LAMSAMA/Akred/M/V/2025
Astronomi	Unggul	10/SK/LAMSAMA/Akred/M/II/2026
Kimia	Unggul	086/SK/LAMSAMA/Akred/M/V/2025
Aktuaria	Unggul	068/SK/LAMSAMA/Akred/M/XII/2022
Sains Komputasi	Unggul	037/SK/LAMSAMA/Akred/M/XII/2022
Pengajaran Matematika	Unggul	218/SK/LAMSAMA/Akred/M/VII/2025
Pengajaran Fisika	Unggul	020/SK/LAMSAMA/Akred/M/II/2026
Pengajaran Kimia	Unggul	554/SK/BAN-PT/Ak.KP/M/II/2024
Ilmu dan Rekayasa Nuklir	Baik (akreditasi pertama)	065/SK/LAMSAMA/Akred/M/VII/2023

Accredited & Certified by:



STATUS AKREDITASI PROGRAM STUDI DOKTOR FMIPA

Program Studi	Akreditasi	No. SK
Matematika	Unggul	128/SK/LAMSAMA/Akred/D/XII/2023
Fisika	Unggul	057/SK/LAMSAMA/Akred/D/VI/2024
Astronomi	Unggul	165/SK/LAMSAMA/Akred/D/VI/2025
Kimia	Unggul	105/SK/LAMSAMA/Akred/D/XI/2023
Rekayasa Nuklir	Baik Sekali (akreditasi pertama)	No. 745/SK/BAN-PT/Ak/D/III/2024

Accredited & Certified by:



QS World University Rankings by Subject of FMIPA ITB

	2023	2024	2025	2026
Chemistry	451-500	551-600	451-500	351-400 #1 domestic
Mathematics	351-400	501-550	401-450	351-400 #1 domestic
Physics and Astronomy	451-500	551 - 600	501-550	401-450 #1 domestic
Natural Sciences	451-500	501-550	401-450	374 #1 domestic



HEAD OF RESEARCH GROUP



Astronomy

Prof. Dr. Dhani Herdiwijaya, M.Sc.
S3 Kyoto University



Nuclear Physics and Biophysics

Prof. Dr. Eng. Ir. Sidik Permana S.Si., M.Eng.
S3 Tokyo Institute Of Technology



Physics of the Earth and Complex Systems

Prof. Dr. Eng. R. Bagus Endar Bachtiar
S3 Kyoto University



High Energy Theoretical Physics

Prof. Dr. rer. nat. Bobby Eka Gunara, S.Si., M.Si.
S3 Martin Luther Universitat



Physics and Technology of Advanced Materials

Prof. Dr. Yudi Darma, S.Si., M.Si.
S3 Hiroshima University Japan



Instrumentation and Computational Physics

Prof. Dr. Eng. Suprijadi, M.Eng.
S3 Nagoya University



Magnetic and Photonic Physics

Prof. Drs. Alexander Agustinus P. Iskandar Ph.D.
S3 University of Durham



Biochemistry and Biomolecular Engineering

Prof. Dessy Natalia Ph.D.
S3 The University of Kent at Canterbury

HEAD OF RESEARCH GROUP



Analytical Chemistry

Prof. Ir. Muhammad Ali Zulfikar M.Si., Ph.D., IPP.
S3 Univ. Kebangsaan Malaysia



Analysis and Geometry

Prof. Iwan Pranoto M.Sc., Ph.D.
S3 University of Toronto



Inorganic and Physical Chemistry

Prof. Dr. Veinardi Suendo S.Si., M.Eng.
S3 Ecole Polytechnique Palaiseau



Combinatorics Mathematics

Prof. Edy Tri Baskoro, M.Sc., Ph.D.
S3 University of Newcastle



Organic Chemistry

Prof. Dr. Lia Dewi Juliawaty M.S.
S3 Chiba University



Industrial and Financial Mathematics

Prof. Dr. Janson Naiborhu S.Si., M.Si.
S3 Institut Teknologi Bandung



Algebra

Prof. Dr. Muchtadi Intan Detiena S.Si., M.Si.
S3 Universite D'Amiens



Statistics

Prof. Udjianna Sekteria Pasaribu, Ph.D.
S3 University of Wales

PROFIL DOSEN



**7 Non Jabatan Fungsi
(3,3%)**



**23 Asisten Ahli
(10,7%)**



**73 Lektor
(34%)**



**46 Lektor Kepala
(21,4%)**



**66 Guru Besar
(30,7%)**

Total 215 Dosen

**11 Dosen Lulusan Magister (S2)
204 Dosen Lulusan Doktor (S3)**



Physics

Physics Department—Key Milestones

Faculteit van Exacte Wetenschap (Faculty of Exact Sciences)

Founded in Bandung as part of the Nood-Universiteit (Emergency University) in Jakarta, with Prof. H. Th. Leeman serving as its first dean.

Faculteit van Wiskunde en Natuurswetenschap

Part of Universiteit Indonesie and named as FIPIA (Fakultas Ilmu Pasti dan Ilmu Alam/*faculty of Exact and Natural Sciences*).

Master and Doctoral Degree in Physics

Master and Doctoral program started.

1947

1948

1950

1959

1979

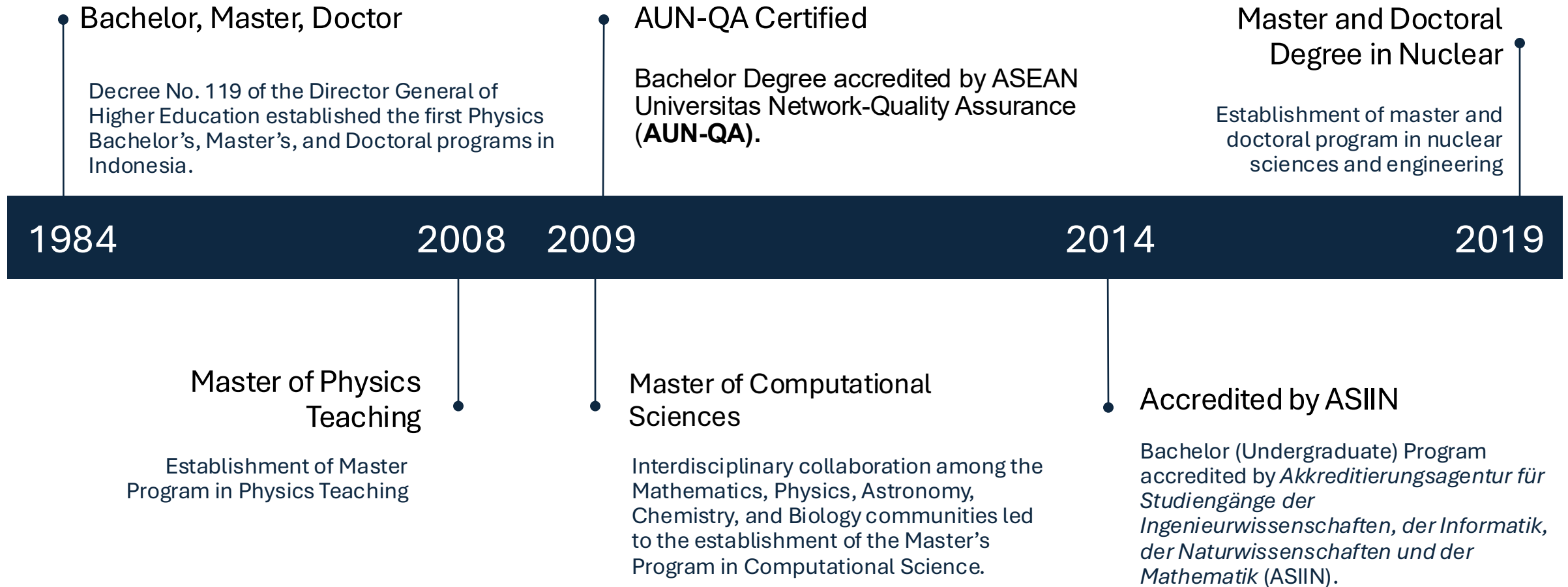
Doktorandus Degree

First institution in Indonesia to confer the Bachelor of Physics (“Doktorandus”, Drs.) degree.

Institut Teknologi Bandung (ITB)

ITB officially established (2 march 1959) and FIPIA becomes one of the founding faculties.

Physics Department—Key Milestones



Physics Department—Achievements

Institutional Awards

- **QUE Program (1998–2003) and PHK-B Program (2005–2007)** supporting the enhancement of undergraduate education quality.
- **Two Research Center Grants** in Electronic Materials Physics and Earth Physics, awarded through ITB's competitive research funding schemes.
- **Numerous research grants** from the Government of Indonesia and ITB, fostering research excellence and innovation.

Individual Awards and Recognition

6

Habibie Awards



- Basic Sciences
- Engineering

5

LIPI Sarwono Award,
Siwabessy Award,
Ahmad Bakrie Award



- Bidang sains
- Bidang kenukliran

2

National Outstanding
Lecturer Award
Recipient(s)

- Education, Research, and
Community Services

50+

Others.

Research & Scientific Excellence,
Institutional & National Services,
Teaching & Academic Leadership,
Innovation & Inventions,
Women in Sciences,
Invited Speakers

Undergraduate Program in Physics (UPP)

Explore Nature. Understand the Universe. Shape the Future.

- ✓ Strong foundation in fundamental and applied physics
- ✓ Integration of theory, experimentation, and computation
- ✓ Research-oriented and innovation-driven learning
- ✓ Development of analytical, quantitative, and problem-solving skills
- ✓ Preparation for careers in research, industry, technology, and academia
- ✓ Pathway to leading graduate programs around the world

Graduates equipped to tackle scientific and technological challenges in a rapidly evolving world.



@physicsitb



www.fi.itb.ac.id



Youtube.com/c/physicsitb



Undergraduate Program in Physics (UPP) — *curriculum*

A curriculum that combines strong scientific foundations, hands-on research experience, and flexible pathways to prepare graduates for advanced studies and diverse careers.

General Studies

34 SKS (≈ 57 ECTS)

Builds essential academic and professional foundations through:

- Mathematics and basic sciences
- Computational thinking
- Sustainability and global perspectives
- Communication skills (Indonesian & English)
- Character and ethical development

Outcome: Strong analytical, communication, and transferable skills.

Core Physics

81 SKS (≈ 135 ECTS)

Develops comprehensive physics competencies through:

- Mechanics, Electromagnetism, Thermodynamics
- Quantum and Statistical Physics
- Mathematical Physics and Computation
- Instrumentation and Experimental Physics
- Research Project / Final Thesis

Outcome: Deep conceptual understanding, problem-solving ability, experimental skills, and research competence.

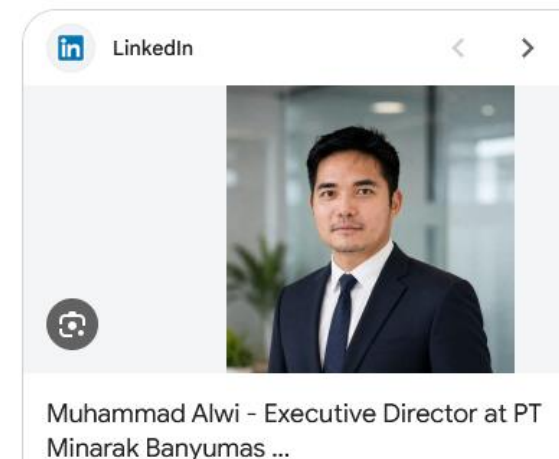
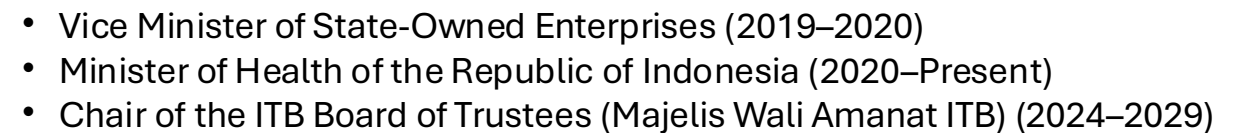
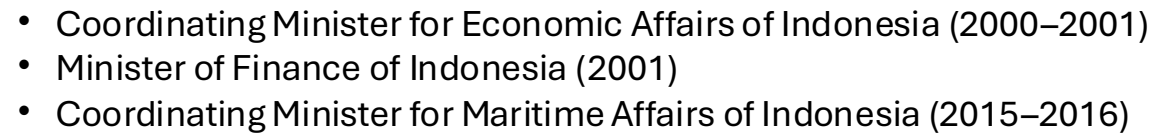
Electives & Experiential Learning

29 SKS (≈ 48 ECTS)

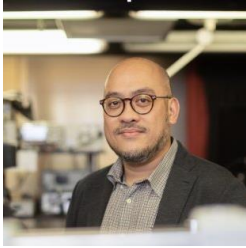
Provides flexibility to pursue individual interests through:

- Advanced physics courses
- Specialization tracks:
 - Energy Physics
 - Medical Physics
 - Natural Resources & Disaster Mitigation
 - Data Science Physics
- Internships, mobility programs, and independent studies

Outcome: Specialized expertise, interdisciplinary perspectives, and career readiness.

[illegible]

Undergraduate Program in Physics (UPP) — *alumni*



Prof. David Marpaung

Fisika 1998

Head of Nonlinear Nanophotonics group
University of Twente, the Netherlands



Dr. Satria Zulkarnaen Bisri

Fisika 2002

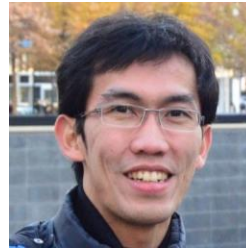
Associate **Professor** at Tokyo University of
Agriculture and Technology



Ahmad Ridwan T. Nugraha

Fisika 2004

Kepala Pusat Penelitian Fisika Kuantum
Badan Riset dan Inovasi Nasional (BRIN)



Andika Putra

Fisika 2006

Senior **Engineering** Technical Writer
MathWorks at MATLAB



Muhammad Anis

Fisika 2008

Staff **Dewan Komisaris PERTAMINA**



Guntur Sianturi

Fisika 2007

Service Delivery Coordinator
Baker Hughes



Randy Budi Wicaksono

Fisika 2009

CEO Ravelware Technology



Ulin Nuha Abdul Qohar

Fisika 2010

Data Scientist at CONSIGLI AS
Norwegia



Prof. David Marpaung

Fisika 1998

Head of Nonlinear Nanophotonics group
University of Twente, the Netherlands



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Fisika 2002

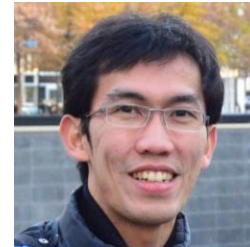
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Norwegia

High Energy Theoretical Physics Research Group

*Exploring the Universe's Deepest Laws
From Gravity to Quantum Realms
Theories Shaping Future Physics*

Powered by Thought, Not Bound by Walls

Our research thrives in the **Theoretical Physics Laboratory**, where ideas transcend space—exploring gravity, quantum fields, and the fabric of the universe from **anywhere**.



Magnetic and Photonic Physics Research Group

*Magnetism and Light in Action
From Theory to Application
Shaping Future Technologies*

Empowering Innovation in Magnetism and Photonics

Our research is backed by state-of-the-art labs for material synthesis, device design, optical testing, and cryogenic studies — enabling breakthroughs from nanomaterials to superconductors.

- Photonic Material Synthesis and Fabrication Laboratory
- Magnetic and Superconductor Materials Laboratory
- Optical Characterization Laboratory
- Photonic Device Design and Calculation Laboratory
- Magnetic & Cryogenic Laboratory



Physics and Technology of Advanced Materials Research Group

Exploring Advanced Functional Materials From Physics to Prototyping Innovating Future Technologies

Pioneering the Future of Materials Science

FTMM's cutting-edge research is powered by advanced laboratories for material synthesis, device fabrication, and computational modeling:

- MOCVD Laboratory (Metal Organic Chemical Vapor Deposition)
- PECVD Laboratory (Plasma Enhanced Chemical Vapor Deposition)
- PLD Laboratory (Pulsed Laser Deposition)
- Sputtering Laboratory
- Nanomaterials Synthesis and Functionalization Laboratory
- Material Computation Laboratory
- Energy and Environmental Materials Laboratory
- Integrated Materials and Instrumentation Laboratory



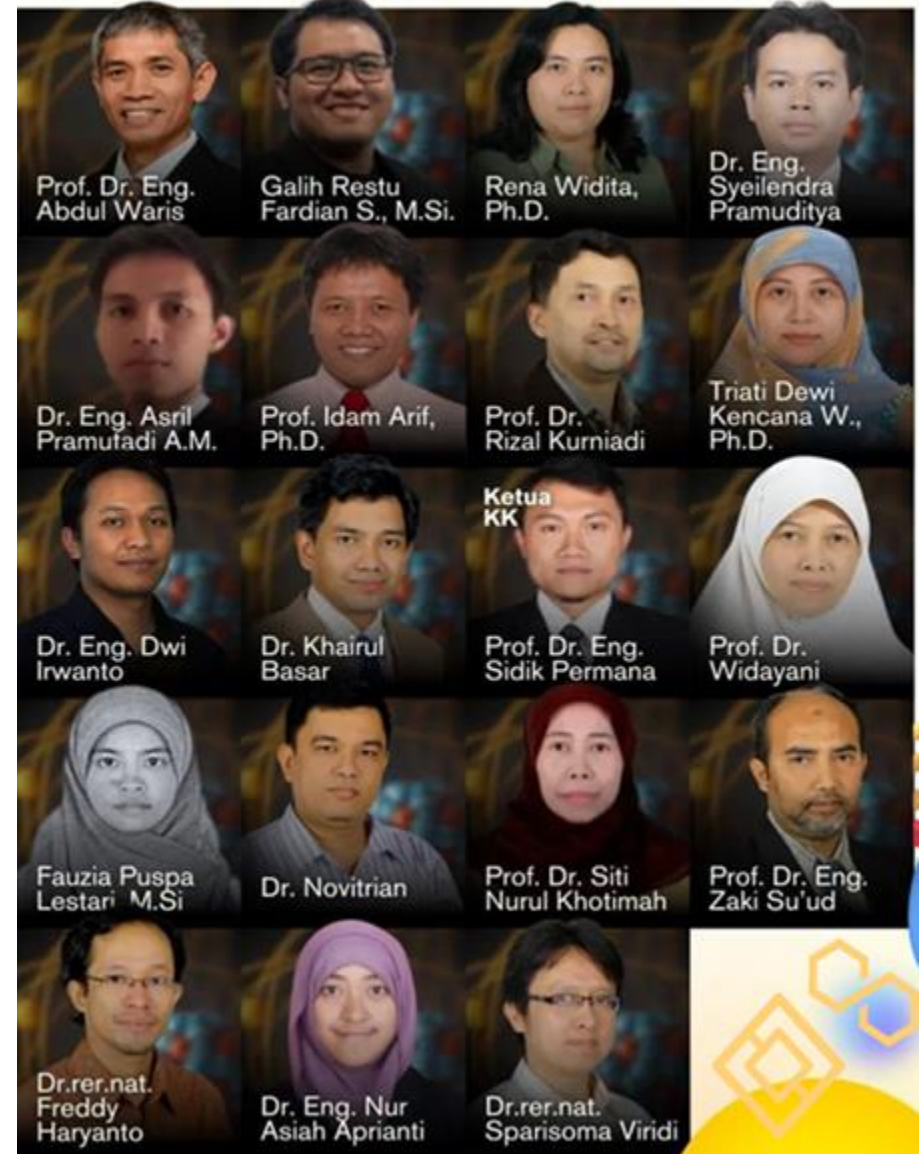
Nuclear Physics and Biophysics Research Group

*From Nuclear Safety to Healing
Quantum, Reactors, and the Brain
Science for Life and Energy*

Advancing Energy and Health Through Nuclear and Biophysical Innovation

Our research is supported by specialized laboratories that enable breakthroughs in reactor design, nuclear safety, medical imaging, and molecular biophysics:

- Nuclear Reactor Physics Laboratory
- Nuclear Computational Laboratory
- Nuclear Fuel Management and Recycling Laboratory
- Biophysics Laboratory
- Medical Physics Laboratory



Earth Physics and Complex Systems Research Group

*Unraveling Earth Through Physics
From Subsurface to Surface Systems
Science for a Resilient Planet*

Exploring Earth's Complexity with Cutting-Edge Science

Our research is powered by advanced laboratories that transform data into deep insights on geophysics, natural systems, and environmental resilience.

- Computational Lab
- Electromagnetic Induction Lab
- Environmental & Disaster Physics Lab
- Geotechnical Earth Physics Lab
- Rock Physics Lab
- Wave & Subsurface Imaging Lab



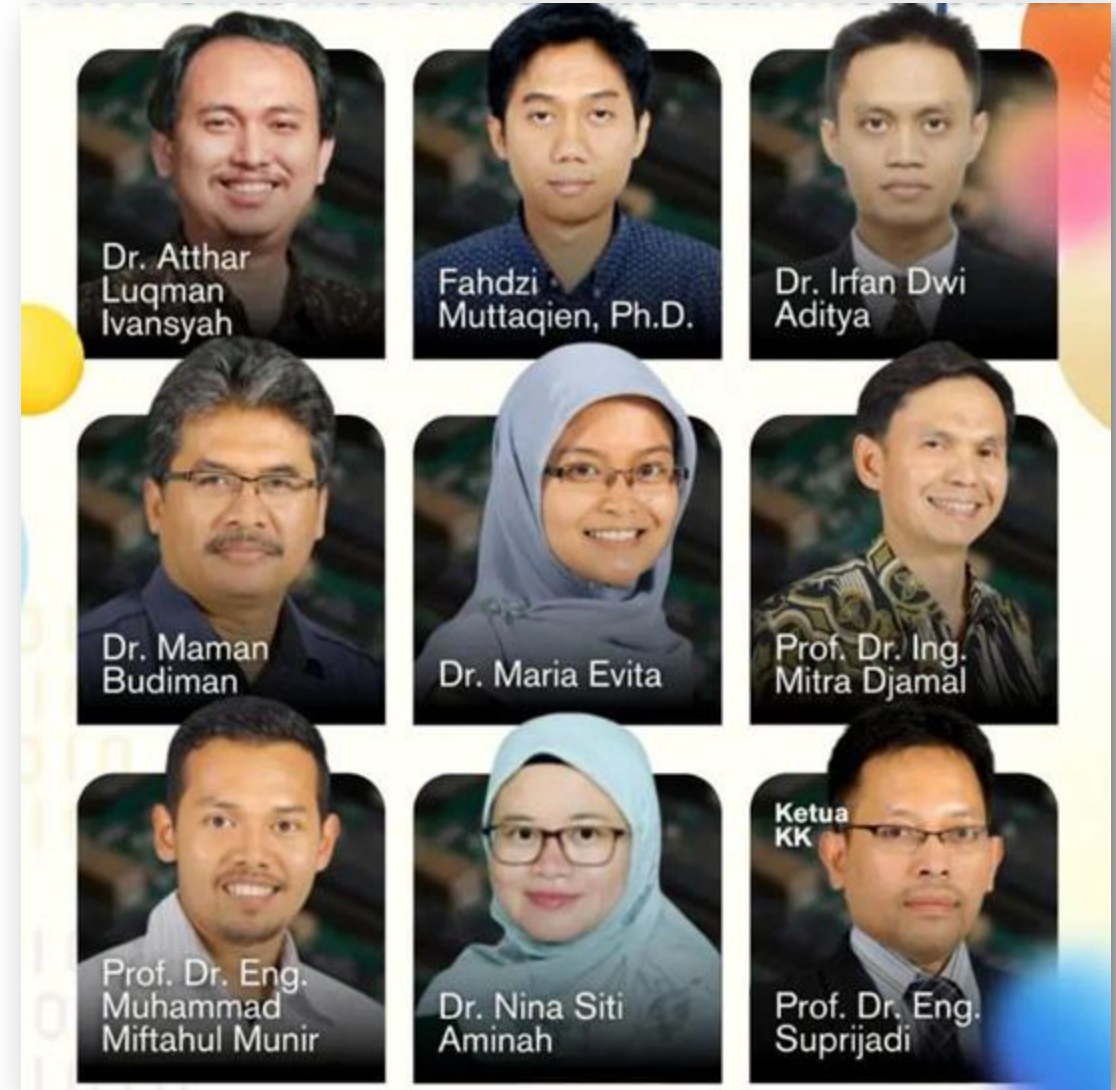
Instrumentation and Computational Physics Research Group

*Smart Sensors, Smarter Systems
From Signals to Solutions
Empowering Science with Innovation*

Driving Innovation Through Smart Instrumentation and Advanced Computing

Our research is supported by specialized laboratories that enable breakthroughs in sensor technology, data processing, and IoT-based systems.

- Sensor Laboratory
- Advanced Computing Laboratory
- Analytical and Aerosol Instrumentation Laboratory (CAAI)
- Network Instrumentation Laboratory (IoT)
- Basic Instrumentation and Optics Laboratory



Three Core Areas for Collaboration

Identifying areas of alignment between industry needs and the educational and research strengths of Physics at ITB.



01

Talent Development & Recruitment

Internships, industrial placements/projects, competency development, and pathways for graduate recruitment.

Focus: measurable outcomes that can be evaluated through jointly designed pilot initiatives.



02

Joint Research

Problem-driven research, measurement and characterization, modelling, computation, prototyping, and validation.

Focus: measurable outcomes that can be evaluated through jointly designed pilot initiatives.



03

Curriculum & Graduate Profile

Understanding the curriculum structure, graduate competencies, and opportunities for alignment with industry requirements.

Focus: measurable outcomes that can be evaluated through jointly designed pilot initiatives.

Graduate Profile: Competencies Relevant to Industry

The program learning outcomes emphasize mastery of physics, modelling, experimentation, communication, effective collaboration, interdisciplinary application, scientific integrity, and lifelong learning.



Physics Foundations

Understand core concepts in classical and modern physics.



Modelling & Analysis

Formulate physical systems as mathematical models and analyse their behaviour.



Computation & Data

Apply mathematical, computational, and information-technology tools.



Experimentation

Design experiments and process, analyse, and interpret experimental data.



Communication & Teamwork

Communicate scientific ideas effectively in oral and written forms and work productively both independently and in teams.



Adaptive & Interdisciplinary

Apply physics across disciplines and continuously develop new knowledge and capabilities.

Examples of professional contribution areas

- Energy & resources**

modelling, geophysics, materials, data
- Materials & devices**

characterization, sensors, semiconductors, photonics
- Instrumentation & automation**

measurement, data acquisition, IoT, control systems
- Information technology**

scientific computing, simulation, data analytics
- Research & postgraduate study**

R&D, laboratories, master's/doctoral study

Value for industry partners: a transferable combination of quantitative analysis, experimentation, and computational capability across sectors.